In [1]:
```
import pandas as pd
dataset=pd.read_csv("Social_Network.csv")
dataset
dataset=pd.get_dummies(dataset,drop_first=True)
print(dataset)
```

```
      User ID  Age  EstimatedSalary  Purchased  Gender_Male
0    15624510   19            19000          0         True
1    15810944   35            20000          0         True
2    15668575   26            43000          0        False
3    15603246   27            57000          0        False
4    15804002   19            76000          0         True
..        ...  ...              ...        ...          ...
395  15691863   46            41000          1        False
396  15706071   51            23000          1         True
397  15654296   50            20000          1        False
398  15755018   36            33000          0         True
399  15594041   49            36000          1        False

[400 rows x 5 columns]
```

In [2]:
```
dataset=dataset.drop(columns=["User ID"])
dataset
```

Out[2]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [15]:
```
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
print(y)
dataset.shape
dataset[["Purchased"]].value_counts()
```

```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         ...
395          1
396          1
397          1
398          0
399          1

[400 rows x 1 columns]
```

Out[15]:

```
Purchased
0            257
1            143
Name: count, dtype: int64
```

In [16]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
```

In [25]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import GridSearchCV
param_grid={
    'penalty':['l1','elasticnet'],
    'solver':['lbfgs','liblinear','newton-cg']
}
reg=GridSearchCV(LogisticRegression(),param_grid,refit=True,verbose=3,n_jobs=-1)
reg.fit(x_train,y_train)
```

```
Fitting 5 folds for each of 6 candidates, totalling 30 fits
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py:489:
FitFailedWarning:
25 fits failed out of a total of 30.
The score on these train-test partitions for these parameters will be set to nan.
If these failures are not expected, you can try to debug them by setting error_score='raise'.

Below are more details about the failures:
--------------------------------------------------------------------------------
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line
851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 1416,
in fit
    solver = _check_solver(self.solver, penalty, self.dual)
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 83, in
_check_solver
    raise ValueError(
    ...<2 lines>...
    )
ValueError: Solver lbfgs supports only 'l2' or None penalties, got l1 penalty.

--------------------------------------------------------------------------------
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line
851, in _fit_and_score
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    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
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  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 83, in
_check_solver
    raise ValueError(
    ...<2 lines>...
    )
ValueError: Solver newton-cg supports only 'l2' or None penalties, got l1 penalty.

--------------------------------------------------------------------------------
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line
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    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
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in fit
    solver = _check_solver(self.solver, penalty, self.dual)
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 83, in
_check_solver
    raise ValueError(
    ...<2 lines>...
```

```
    )
ValueError: Solver lbfgs supports only 'l2' or None penalties, got elasticnet penalty.

--------------------------------------------------------------------------------
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line
851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 1416,
in fit
    solver = _check_solver(self.solver, penalty, self.dual)
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 91, in
_check_solver
    raise ValueError(
        f"Only 'saga' solver supports elasticnet penalty, got solver={solver}."
    )
ValueError: Only 'saga' solver supports elasticnet penalty, got solver=liblinear.

--------------------------------------------------------------------------------
5 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line
851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 1416,
in fit
    solver = _check_solver(self.solver, penalty, self.dual)
  File "C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py", line 83, in
_check_solver
    raise ValueError(
    ...<2 lines>...
    )
ValueError: Solver newton-cg supports only 'l2' or None penalties, got elasticnet penalty.

  warnings.warn(some_fits_failed_message, FitFailedWarning)
C:\Users\User\anaconda3\Lib\site-packages\sklearn\model_selection\_search.py:1234: UserWarning:
One or more of the test scores are non-finite: [   nan 0.8125    nan    nan    nan    nan]
  warnings.warn(
C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py:1403: FutureWarning:
'penalty' was deprecated in version 1.8 and will be removed in 1.10. To avoid this warning, leave
'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of
penalty='l2', l1_ratio=1 instead of penalty='l1', l1_ratio set to a float between 0 and 1 instead
of penalty='elasticnet', and C=np.inf instead of penalty=None.
  warnings.warn(
C:\Users\User\anaconda3\Lib\site-packages\sklearn\linear_model\_logistic.py:1429: UserWarning:
Inconsistent values: penalty=l1 with l1_ratio=0.0. penalty is deprecated. Please use l1_ratio
only.
  warnings.warn(
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to
(n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[25]:

▸ GridSearchCV

▸ best_estimator_: LogisticRegression

▸ LogisticRegression

In [26]:
```
y_pred=reg.predict(x_test)
print(y_pred)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_pred,y_test)
print(cm)
```

```
[0 1 0 1 0 0 1 0 0 0 0 1 0 0 0 0 1 1 0 1 0 0 0 1 0 0 1 0 1 0 0 0 1 0 1 0 0
 0 0 0 0 0 0 0 0 1 0 0 1 0 0 1 0 0 0 0 1 0 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
 0 0 1 1 0 0]
[[50  8]
 [ 2 20]]
```

In [27]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.96      0.86      0.91        58
           1       0.71      0.91      0.80        22

    accuracy                           0.88        80
   macro avg       0.84      0.89      0.85        80
weighted avg       0.89      0.88      0.88        80
```

In [28]:
```
from sklearn.metrics import r2_score
r=r2_score(y_pred,y_test)
print(r)
```

```
0.37304075235109724
```

In [ ]:

In [ ]: